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Unit “ 4 : Session “ 4 : SLO “ 1 | Branch Instructions “ Choose the Best

■ I. Multiple Choice

1. c) Arithmetic branch “ not a branch type; branches are conditional, unconditional, or call/return.
2. b) Zero flag “ most conditional branches (BEQ/BNE) test the zero flag.
3. b) JMP (Jump) “ an unconditional jump permanently alters the flow; conditional branches may fall through.
4. a) Program counter increments normally “ with carry = 0 the BCS condition fails, so the branch is not taken and execution falls through.
5. b) Relative “ branch instructions typically use PC-relative addressing so code stays position-independent.
6. c) To save the return address and jump “ a subroutine call pushes the return address (stack/link register) before transferring control.
7. c) BMI (Branch if Minus) “ it tests the sign/negative flag, so it is conditional; JMP/BRA/JSR are unconditional.
8. b) It stalls “ a taken branch flushes wrongly prefetched instructions, causing pipeline bubbles.
9. a) Branch prediction “ speculatively fetching the likely path reduces branch penalties.
10. a) BPL (Branch if Plus) “ a conditional branch back to the loop start creates a loop; RTS returns, NOP/HLT do not loop.